Savitribai Phule Pune University

A PROJECT REPORT ON

Simulation Of Quantum Harmonic Oscillator and applications

Submitted by

Ugale Hritik Ganesh

M.Sc II

Under the Guidance of

Prof S.P GUPTA

Department of Physics

H. P. T. Arts and R. Y. K. Science College,

Nashik-05

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CERTIFICATE

This is certifying that Mr. Ugale Hritik Ganesh of M.Sc. Physics, Savitribai **Phule Pune** University Exam seat No. is a bonafied student of this college. He has satisfactorily completed project work entitled **"Simulation of Quantum Harmonic Oscillator and applications"** Completion of project course for the Degree of M.Sc. (Physics) of Savitribai Phule Pune University, Pune During the year 2023-2024.

Prof . S.P Gupta
(Project Guide)

Prof. Dr. M. D. Deshpande
(Head of Department)

Internal Examiner

External Examiner

ACKNOWLEDGEMENT

With the immense pleasure, I am presenting this report as a fulfillment for the degree Masters of Science in Physics of the **Savitribai Phule Pune University, Pune**.

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I am very thankful to peoples who directly or indirectly involved in my preparatory work. .

Ugale Hritik Ganesh

MATLAB code for Hydrogen atom

1

```
% qp_h_pot.m clear all
close all clc
% Inputs --------------------------------------------------------------- L = 2;             % Orbital
quantum number
num = 5001;        % Number of data points r_min = 1e-12;
r_max = 25e-10;
% Constants ------------------------------------------------------------ hbar = 1.055e-34;     %
J.s
e = 1.602e-19;        % C me = 9.109e-31;        % kg eps0 = 8.854e-12;     % F/m
% Setup ----------------------------------------------------------------- % radial position
r = linspace(r_min,r_max,num);
% Calculate potnetial energies
U_c = -(e/(4*pi*eps0))./r;            % Coulomb interaction
U_L = (hbar^2*L*(L+1)/(2*me*e))./r.^2; % Angular momentum
U_eff = U_c + U_L;                    % Effective potential energy
% Graphics -------------------------------------------------------------- % black & white plot
figure(1)
set(gcf,'Units','Normalized')
set(gcf,'Position',[0.2 0.15 0.2 0.2])
set(gca,'fontsize',8);
numS = 200:1000:num; plot(r(numS),U_c(numS),'ko') hold on
plot(r(numS),U_L(numS),'ks') plot(r,U_eff,'k','LineWidth',2)
h_L = legend('\itU_c', '\itU_L', '\itU_{eff}','Orientation','horizontal'); set(h_L,'Box','off');
```

```
grid on
plot(r,U_c,'k') plot(r,U_L,'k')
axis([r_min r_max -20 20]) tm1 = '{\itL} = ';
tm2 = num2str(L,1);
tm = [tm1 tm2];
title(tm,'FontSize',12);
xlabel('radial position  {\itr}  (m)','FontSize',12); ylabel('potential energy  {\itU}
(eV)','FontSize',12')
% colour plot
figure(2)
set(gcf,'Units','Normalized')
set(gcf,'Position',[0.5 0.15 0.2 0.2])
set(gca,'fontsize',8);
plot(r,U_c,'b') hold on
plot(r,U_L,'r')
plot(r,U_eff,'k','LineWidth',2) axis([r_min r_max -20 20])
h_L = legend('\itU_c', '\itU_L', '\itU_{eff}','Orientation','horizontal'); set(h_L,'Box','off');
tm1 = '{\itL} = ';
tm2 = num2str(L,1);
tm = [tm1 tm2];
title(tm,'FontSize',12);
xlabel('radial position  {\itr}  (m)','FontSize',12); ylabel('potential energy  {\itU}
(eV)','FontSize',12') grid on
```

2

```
% Constants
c = 2.99e8;           % speed of light (m/s)
h = 6.6256e-34;       % Planck's constant (J.s)
 hbar = 1.055e-34;    % Planck's constant hbar (J.s)
 e = 1.60210e-19;     % electron charge (C)
me = 9.109e-31;
% electron mass (kg) mp = 1.6726e-27;
% mass of proton (kg) mn = 1.6749e-27;
eps0 = 8.8543e-12;    % permittivity of free space (F/m)
 % Inputs
Z = 1;                % atomic number
m = mp;               % hydrogen - 1
mr = m * me / (m + me); mr = me;
E0 = - mr * Z^2 * e^4 / (8 * eps0^2* h^2); a0 = h^2 * eps0 / (pi * me * e^2);

% Outputs
n = 1: 10;
Rn = (n.^2 .* a0) .* 1e10 ./Z; En = (E0 ./ n.^2) ./ e;
Rn(1) En(1)
disp('     n     Rn     En')
disp([n; Rn; En]')
```

3

```
% qp_hydrogen.m
% calls simpson1d.m to find area under curve
clear close all clc
% INPUTS
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
disp('   ');
disp('Solution of the [1D] Radial Schrodinger Equation for hydrogen like atoms')
disp('    ')
r_max = input('max radial distance (default 10e-10 m), r_max =  '); disp('   ');
L = input('orbital quantum number (default 0), L =  '); disp('   ');
m_L = input('magnetic quantum number (default 0), m_L =  '); disp('   ');
Z = input('nuclear charge (default 1), Z =  '); disp('    ');
disp(' Please wait: calculating ....');
num = 1201;                           % number of data points default 1201 % CONSTANTS
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
hbar = 1.055e-34;      % J.s e = 1.602e-19;         % C me = 9.109e-31;        % kg eps0 =
8.854e-12;      % F/m
% POTENTIAL WELL
+++++++++++++++++++++++++++++++++++++++++++++++++++++++ % potential
energy in electron volts (eV)
r_min = 1e-15;    % default 1e-15 r = linspace(r_min,r_max, num); dr = r(2)-r(1);
dr2 = dr^2;
% Coulomb term
K = -Z*e/(4*pi*eps0); U_c = K./r;
% Angular momnetum term
U_L = (hbar^2*L*(L+1)/(2*me*e))./r.^2;
% Effective potential energy
U = U_c + U_L;
U(U < -2000) = -2000;     % default value 1000 U(U > 1000) = 1000;
U_matrix = zeros(num-2,num-2); for cn = 1:(num-2)
   U_matrix(cn,cn) = U(cn+1);
end
tic;
```

```
% SOLVE SCHRODINGER EQUATION
++++++++++++++++++++++++++++++++++++++++++++++ % Make Second Derivative
Matrix
off     = ones(num-3,1);
SD_matrix = (-2*eye(num-2) + diag(off,1) + diag(off,-1))/dr2;
% Make KE Matrix
K_matrix = -hbar^2/(2*me*e) * SD_matrix;
% Make Hamiltonian Matrix
H_matrix = K_matrix + U_matrix;
% Find Eignevalues E_n and Eigenfunctions psi_N
[e_funct, e_values] = eig(H_matrix);
% All Eigenvalues 1, 2 , ... n  where E_N < 0
flag = 0;
n = 1;
while flag == 0
   E(n) = e_values(n,n); if E(n) > 0, flag = 1; end % if
   n = n + 1; end  % while E(n-1) = []; n = n-2;
psi  = zeros(num,n); prob = zeros(num,n);
% Corresponding Eigenfunctions 1, 2, ... ,n: Normalizing the wavefunction for cn = 1 : n
psi(:,cn) = [0; e_funct(:,cn); 0];
area = simpson1d((psi(:,cn).* psi(:,cn))',r_min,r_max); psi(:,cn) = psi(:,cn)/sqrt(area);
prob(:,cn) = psi(:,cn) .* psi(:,cn); end % for
% CALCULATE EXPECTATION VALUES
++++++++++++++++++++++++++++++++++++++++++++
disp('  ')
disp('Enter Principal Quantum Number for calculation of')
disp('   expectation values and graphical display')
pqn = input('Enter Principal Quantum Number (n > L),  n  =  '); qn = pqn-L;
bra = psi(:,qn)'; Psi = psi(:,qn)';
Prob_density = Psi .* Psi;
% probability
ket = Psi;
```

```
braket = bra .* ket;
Prob = simpson1d(braket,r_min,r_max);
% position r
ket = r .* Psi; braket = bra .* ket;
ravg = simpson1d(braket,r_min,r_max);
% position of prob_density peak
r_peak = r(Prob_density == max(Prob_density));
%position r^2
ket = (r.^2) .* Psi; braket = bra .* ket;
r2avg = simpson1d(braket,r_min,r_max);
% momentum ip
ket = gradient(Psi,r)*hbar; braket = bra .* ket;
ipavg = simpson1d(braket,r_min,r_max);
% momentum ip^2
psi1 = gradient(Psi,r);
ket = -gradient(psi1,r)*hbar^2; braket = bra .* ket;
ip2avg = simpson1d(braket,r_min,r_max);
%potential energy U
ket = U .* Psi; braket = bra .* ket;
Uavg = simpson1d(braket,r_min,r_max);
%kinetic energy K
psi1 = gradient(Psi,r);
ket = - (hbar^2/(2*me*e))*gradient(psi1,r); braket = bra .* ket;
Kavg = simpson1d(braket,r_min,r_max);
%total energy
psi1 = gradient(Psi,r);
ket = - (hbar^2/(2*me*e))*gradient(psi1,r) + (U.*Psi); braket = bra .* ket;
Eavg = simpson1d(braket,r_min,r_max);
% Calculate uncertainites
deltar = sqrt(r2avg - ravg^2); deltaip = sqrt(ip2avg + ipavg^2); drdp = (deltar *
deltaip)/hbar;
E_max = max(abs(E));
```

```
% DISPLAY VALUES
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
t_execution = toc; clc
disp('-------------------------------------------------------------------'); fprintf('No. bound states
found =  %0.0g   \n',n)disp('  ');
disp('Quantum State / Eigenvalues  En  (eV)'); for cn = 1 : n
    fprintf('  %0.0f   ',L+cn);     fprintf('   %0.5g   \n',E(cn)); end
disp('  ');
fprintf('Principal Quantum Number, n  =  %0.0g   \n',pqn);
fprintf('Orbital Quantum Number, L  =  %0.0g   \n',L);
fprintf('Magnetic Quantum Number, m_L  =  %0.0g   \n',m_L);
fprintf('Nuclear charge, Z  =  %0.0g   \n',Z); disp('   ')
fprintf('Energy, E =  %0.6g   \n',E(qn)); fprintf('Total Probability = %0.6g   \n',Prob); disp('  ')
fprintf('r_peak = %0.6g   m\n',r_peak);
fprintf('<r> = %0.6g   m\n',ravg);
fprintf('<r^2> = %0.6g   m^2\n',r2avg);
disp('  ')
fprintf('<ip> = %0.6g   N.s\n',ipavg); fprintf('<ip^2> = %0.6g   N^2.s^2\n',ip2avg); disp('  ')
fprintf('<U> = %0.6g   eV\n',Uavg);
fprintf('<K> = %0.6g   eV\n',Kavg);
fprintf('<E> = %0.6g   eV\n',Eavg);
fprintf('<K> + <U> = %0.6g   eV\n',Kavg+Uavg); disp('  ')
fprintf('deltar = %0.6g   \n',deltar);
fprintf('deltaip = %0.6g   \n',deltaip);
fprintf('(dr dk)/hbar = %0.6g   \n',drdp); disp('  ')
fprintf('execution time = %0.3g s  \n',t_execution);
% GRAPHICS ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++ %
plot potential energies
figure(1);
  set(gcf,'Name','Potentials','NumberTitle','off')
  set(gcf,'Units','normalized','Position',[0.44 0.62 0.2 0.3]);   plot(r.*1e9,U_c,'b','linewidth',2);
  hold on;
  plot(r.*1e9,U_L,'r','linewidth',2);   plot(r.*1e9,U,'k','linewidth',2);   legend('coul', 'Ang Mon', 'eff');   grid on;
  axis([1e9.*r_min 1e9.*r_max -5*E_max 5*E_max]);   xlabel('radial position  r  (nm)');
  ylabel('potential energy  U  (eV)');
  title('Potential Energy Functions','fontweight','normal');   set(gca,'fontsize',12')
% Plot Potential Energy and eigenvalues -----------------------------
```

ABSTRACT

The above project focuses on the simulation of the quantum harmonic oscillator using MATLAB. The quantum harmonic oscillator is a fundamental model in quantum mechanics, describing the behavior of a particle in a harmonic potential well The MATLAB scripts are used to give the solution of the Schrodinger Equation for a variety of potential energy functions using a matrix method where the solution are the eigenvalues and Eigen functions of the energy operator. Solving the one-dimensional Schrodinger Equation for bound states in a variety of potential wells using a matrix method that evaluates the eigenvalues and Eigen functions. Solving the one-dimensional Schrodinger Equation for bound states in a variety of potential wells using a matrix method that evaluates the eigenvalues and Eigen functions. The MATLAB (matrix-lab oratory) programming environment is especially useful in conveying these concepts to students because it is geared towards the type of matrix manipulations useful in solving introductory quantum physics problems

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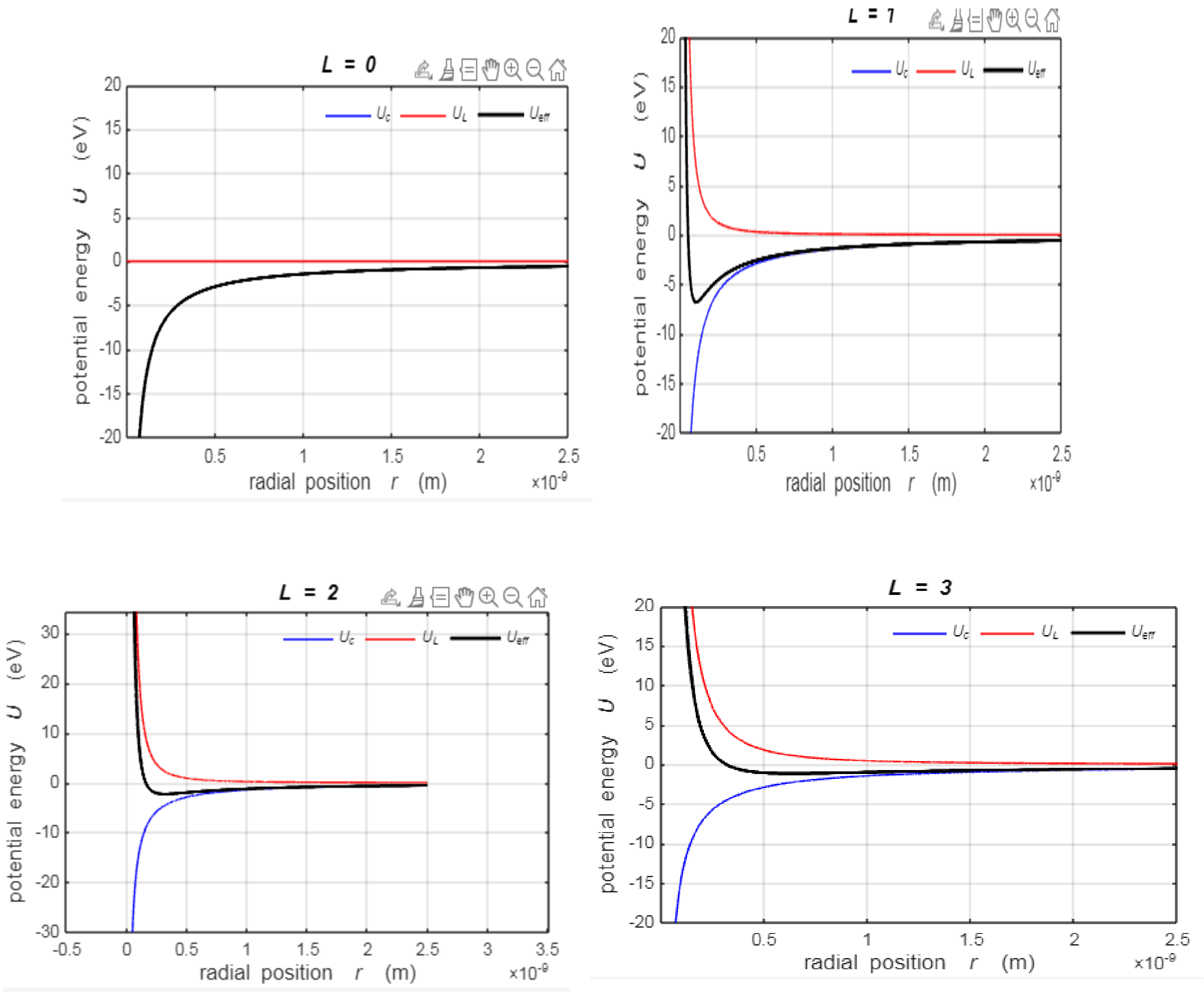
Objectives

1) Simulating Quantum Harmonic Oscillator using MATLAB
2) Making a tool to visualize behavior of particle for better clarity.
3) Studying other Quantum applications such as potential Hydrogen atom and model them
4) Generating Potential wells with user defined data.
5) Solving the one-dimensional Schrodinger Equation for bound states of Hydrogen Atom.

Hydrogen Atom Generated Plots

The Matlab mscript qp_pot.m can be used to plot the potential energy functions as



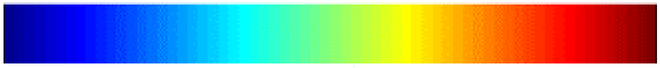
Plots of the potential energy functions Ueff, Uc and Ul for l = 0, 1, 2, 3

The theoretical values for E are calculated from equation (8) using the mscript qp_bohr.m .

	H Bohr	
n	$\boldsymbol{E_B}$**(eV)**	$\boldsymbol{r_{n(*10^{-10})m}}$
1	13.605	0.5292
2	3.401	2.112
3	1.512	4.763
4	0.850	8.467
5	0.5442	13.2295
6	0.3779	19.0505
7	0.2776	25.9299
8	0.2126	33.8676
9	0.1680	42.8637
10	0.1360	52.9182

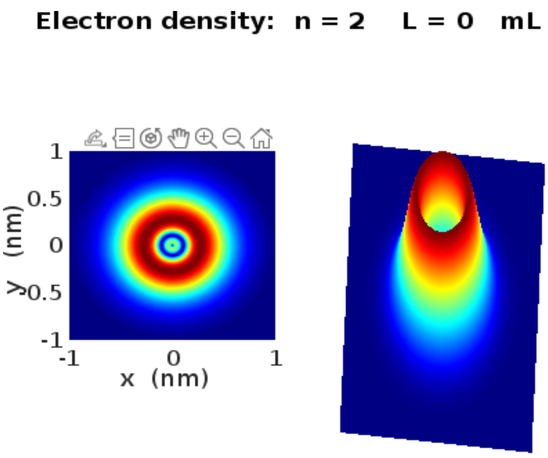
Two-dimensional representation of the probability density functions.

Probability

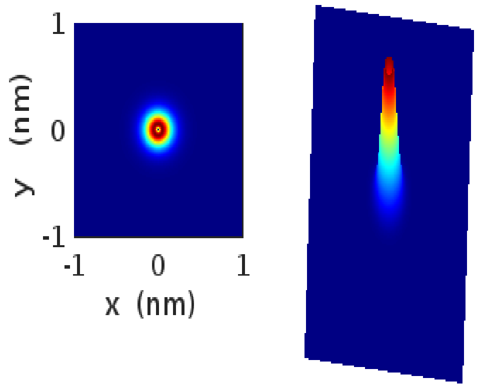


Low ⟶ High

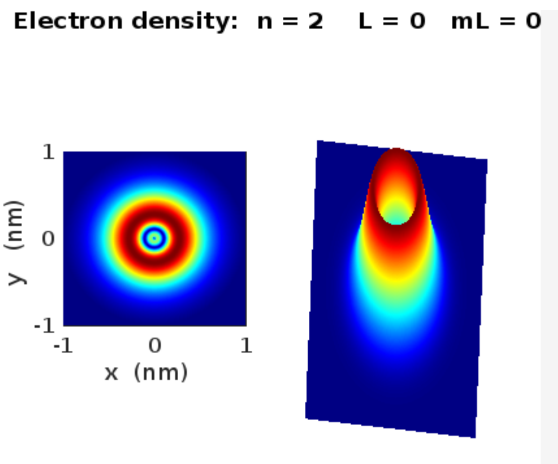
State 1s For *n=2 ,l=0,m=0*



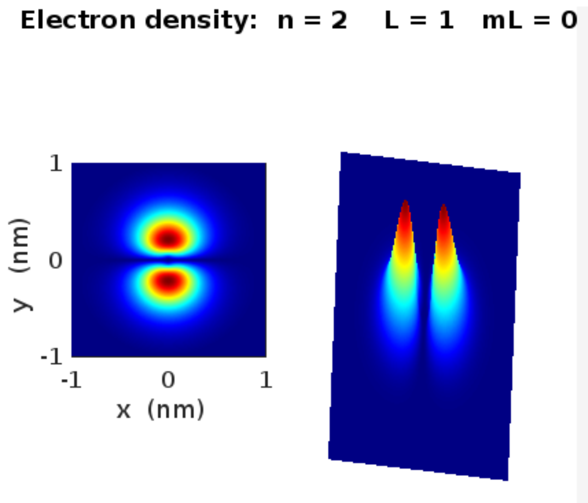
state 1s n=1,l=0,m=0



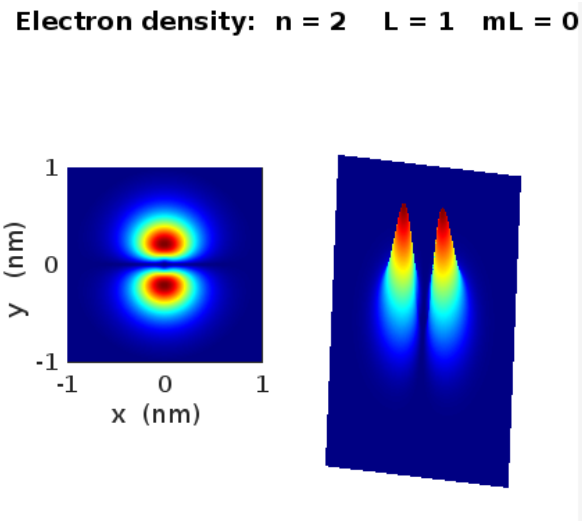
state 2s n = 2 l = 0 ml = 0



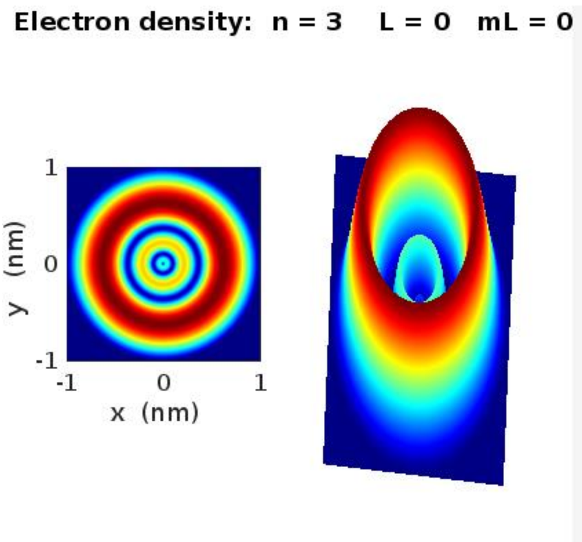
state 2p n = 2 l = 1 ml = 0



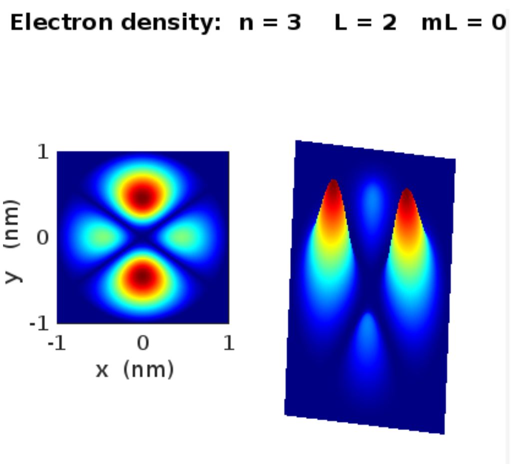
state 2p n = 2 l = 1 ml = 0



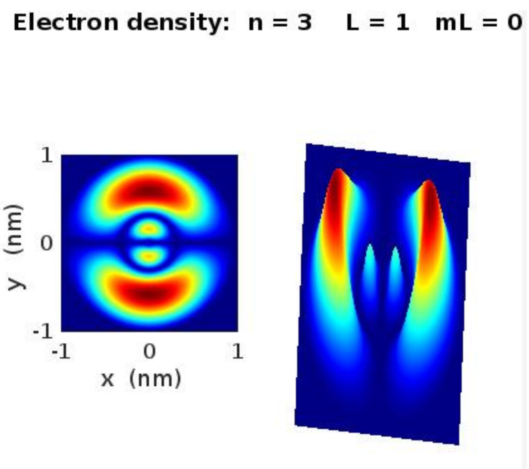
state 3s n = 3 l = 0 ml = 0

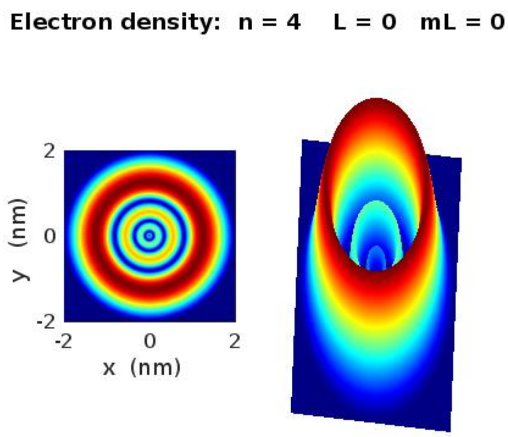


state 3p n = 3 l = 1 ml = 0



state 3d n = 3 l = 2 ml = 0





Introduction

The quantum harmonic oscillator describes behavior of particle in a potential well with a quadratic potential energy function The potential energy function resembles that of a classical harmonic spring mass system Provides insights of energy quantization and wave functions The oscillator properties are applicable to wide range of systems such as atom molecules MATLAB provides powerful tools for visualizing the wave function and probability density of the quantum harmonic oscillator. Plots can be created to display the real and imaginary parts of the wave function, as well as the probability density, which gives the likelihood of finding the particle at a given position.

Parameter Tuning: Users can adjust parameters such as mass, angular frequency, grid size, and domain length to explore different scenarios and observe their effects on the wave function and energy spectrum. Interactive features can be added to allow users to dynamically modify simulation parameters and visualize the results in real-time.

Applications: Simulations of the quantum harmonic oscillator have numerous applications in physics research, quantum computing, nanotechnology, material science, and more.

Understanding the behavior of quantum systems, predicting experimental outcomes, and designing quantum algorithms are among the key applications**.**

Further Extensions. Once the basic simulation is implemented, it can be extended to more complex scenarios, such as time-dependent potentials, multi- dimensional systems, or coupled oscillators. Advanced techniques like density functional theory (DFT) or quantum Monte Carlo (QMC) methods can also be explored for more accurate simulations. In summary, simulating the quantum harmonic oscillator using MATLAB provides a valuable tool for studying quantum mechanics and exploring its applications across various fields. With its versatility and computational capabilities, MATLAB enables researchers and students to gain insights into the behavior of quantum systems and develop innovative solutions to real-world problems

The quantum harmonic oscillator (QHO) is a fundamental piece of physics. Many subjects converge in the study of the QHO, among which modern physics, quantum chemistry, condensed matter [structure of matter and quantum mechanics stand out. Understanding the QHO solution provides the student with powerful tools to tackle more complex problems related to quantum physics. Furthermore, the QHO can model and explain practical phenomena, including infrared and Raman spectroscopies blackbody radiation , stereo mutation in chiral molecules and entropy calculations of single molecules.

In nature, a quantum harmonic oscillator is a micro-physical system with a mathematical structure derived with the help of Bohr correspondence principle from the classical harmonic oscillator. There are numerous micro-physical systems like this. The diatomic molecule provides an example, as long as the energy is not too high and there is no rotation. It is frequently much easier to evaluate a system like a diatomic molecule in terms of how it moves as a whole—such as vibrate or rotate—rather than breaking it down into its numerous constituent particles. Collective motion refers to the overall motion of the system. The transition from classical to quantum entities is accomplished by substituting classical quantities such as position x and momentum p with the corresponding quantum mechanical operators Q and P, as per the quantization principles. The Hamiltonian H, a quantum energy operator, is then derived from the classical energy formula **$E=p^2/2\mu+1/2kx^2$.**

We can solve the Schrodinger Equation for a variety of different potential energy functions. The m-script Sewells_1.m defines most of the constants and potential well parameters. Within the m-script you can change the code to modify the potential wells and add new potential wells. The first step in solving the Schrodinger Equation using the Matrix Method for a bound electron is to run the file Sewells_1.m and select the type of potential well. We can see how the time-independent Schrodinger Equation in one dimension is plausible for a particle of mass m, whose motion is governed by a potential energy function U(x) by starting with the classical one dimensional wave equation and using the de Broglie relationship.

In the quantum simple harmonic oscillator, a particle of mass ‘m’ is confined within a potential well created by a spring with spring constant ‘k’. The classical simple harmonic oscillator obeys Hooke’s law, and its potential energy is given by The energy levels of the quantum harmonic oscillator are

$$E=(n+\frac{1}{2})h/2\pi w$$

n=0,1,2,… is a non-negative integer corresponding to the energy level of the oscillator.

The next section of my project involves plotting the different types of potential well which are user defined using simple commands thereby understanding the well topologies.

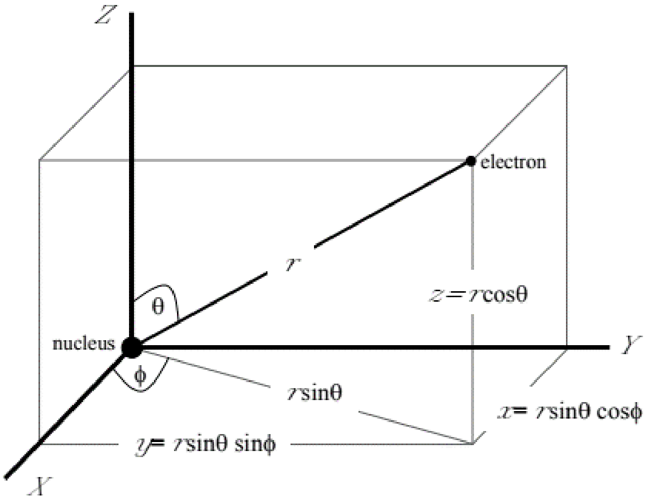
The later section of the above project focuses on solving the Schrodinger equation for 1 electron system that is the Hydrogen atom problem by plotting the potential energy functions for different orbital quantum numbers and simulation of probability density functions and comparing the result with the theoretical calculations

Hydrogen Atom

Hydrogen is the simplest of all the atoms with only one electron surrounding the nucleus. the mass of the electron is much less the nuclear mass, therefore, we will assume a stationary nucleus exerting an attractive force that binds the electron to the nucleus. This is the Coulomb force with corresponding potential energy Uc(r) is

$$\text{Uc(r)}=\frac{Ze^2}{4\pi\varepsilon 0 r}$$

The Coulomb force between the nucleus and electron is an example of a central force where the attractive force on the electron is directed towards the nucleus. This is a three-dimensional problem and it best to use spherical coordinates. (r,θ,φ)) centered on the nucleus. The radial coordinate is r,θ is the polar angle (0 to π) and φ is the azimuthal angle (0 to 2 π)



The time independent Schrodinger Equation in spherical coordinates can be expressed as

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial\Psi}{\partial r})+\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}\left(sin\theta\frac{\partial\Psi}{\partial r}\right)+\frac{1}{r^2 sin\theta^2}\frac{\partial^2\Psi}{\partial\varphi^2}+2\frac{\mu}{\hbar^2}(E-U)\Psi=0 \qquad (1)$$

where μ is the reduced mass of the system

$$\mu=\frac{m_e m_p}{m_e+m_p}$$

The time independent wave function Ψ in spherical coordinates is given by

$\Psi(r,\theta,\varphi)$=R(r)$\Theta(\theta)\emptyset(\varphi)$ (2)

Equation (1) is separable, meaning a solution may be found as a product of three functions, each depending on only one of the coordinates .(r,θ,φ))

This substitution allows us to separate equation 1 into three separate differential equations
Such as

$$\frac{d^3\Phi(\varphi)}{d^2\varphi} = -m^2\Phi(\varphi)$$ **azimuthal equation**

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{\Theta(\theta)}{d\theta}\right)+\left(l(l+1)-\frac{m_l^2}{\sin^2\theta}\right)\Theta(\theta)=0$$

angular equation

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2m}{\hbar^2}\left(E-U_c-\frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right)R=0$$

radial equation

the angular equation (equation 6) depends upon the quantum numbers m and l
The solution of the angular equation was first worked out by the famous mathematician Adrien Legendre (1752 – 1833). Angular Equation is often called the associated Legendre equation.

The solutions $\Theta(\theta)$ for the angular equation are polynomials in cosθ known as the associated Legendre polynomials $pl^{ml}(cos\theta)$
where *l=0,1,2…..and m=0,-+* 1,2,3…
The letters arose from visual observations of spectral lines: sharp, principle, diffuse, and fundamental. After l = 3 (f state), the letters generally follow the order of the alphabet. Atomic states are normally referred to by the number n and the l letter.

The normalized solution to angular equation is

$$\Theta(\theta) = N_{lm_l} P_l^{m_l}(\cos\theta)$$

It is customary for historical reasons to use letters for the various values of l

l	0	1	2	3	4	5
letter	s	p	d	f	g	h

The differential equations in φ azimuthal equation and radial equation are independent of the potential energy function Uc(r) The total energy E and the potential energy Uc(r) appear only in the radial differential equation. Therefore, it is only the radial equation containing the potential energy term Uc(r) that determines the allowed values for the total energy E.

Physically acceptable solutions of the radial equation

$$E_n = \frac{z^2 m_e 4}{(4\pi\varepsilon 0)^2 2\hbar^2}\frac{1}{n^2} = -13.6\frac{z^2}{n^2}eV$$ **total energy is quantized.**

where the principal quantum number is n = 1, 2, 3, … and n > l. The negative sign indicates that the electron is bound to the nucleus. If the energy were to become positive, then the electron would no longer be a bound particle and the total energy would no longer be quantized. The quantized energy of the electron is a result of it being bound to a finite region. The energy levels of the hydrogen atom depend only on the principle quantum number n and do not depend on any angular dependence associated with the quantum numbers l and ml. above equation is in agreement with the predictions of the Bohr model. In the Bohr Model of the atom the total energy En is quantized and the electron can only orbit without radiating energy in stable orbits of fixed radii *rn* given by equation.

$$r_n = \frac{4\pi\varepsilon 0}{\mu e^2}\hbar^2 n^2$$ **Bohr model allowed state orbits**

For hydrogen-like species, the total energy depends only on the principal quantum number n, this is not the case for more complex atoms. The ground state is specified by the unique set of quantum n = 1, l = 0, ml = 0. For the first excited state there are four independent wave functions with quantum numbers:

$n = 2 \quad l = 0 \quad m_l = 0$
$n = 2 \quad l = 1 \quad m_l = -1$
$n = 2 \quad l = 1 \quad m_l = 0$
$n = 2 \quad l = 1 \quad m_l = 1$

his means that the first excited state is four-fold degenerate as the total energy E2 only depends on the principle quantum number n. We can define a pseudo-wave function *g(r)=rR(r)* which leads to a one dimensional Schrodinger Equation

$$-\frac{\hbar^2}{2m}\frac{d^2 g(r)}{d_{r^2}} + Ueff(r)g(r) = E_g(r)$$

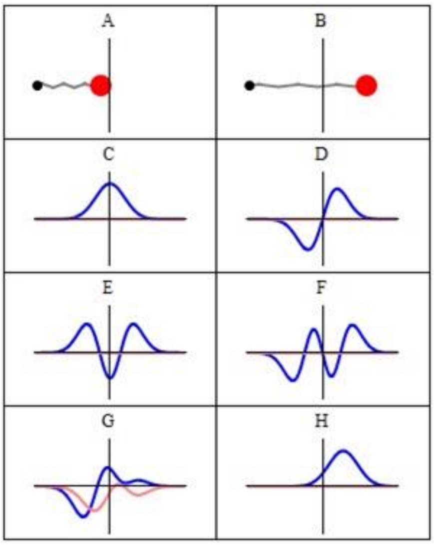
where the effective potential energy $Ueff$ has two contributions due to the Coulomb interaction *Uc* and the angular motion of the electron *Ul*

$Ueff = Uc + Ul \qquad Uc = \frac{Ze^2}{(4\pi\varepsilon 0)r} \qquad Ul = \frac{l(l+1)}{2m^2}$

Analytical solution of Quantum Harmonic Oscillator

The QHO can be used to describe almost any system in which a certain entity makes small vibrations with respect to an equilibrium point. To obtain a full-fledged analytical solution for the QHO an eight step solution can be implemented he sum of these eight small problems constitutes the analytical solution for QHO. These sections are as follows:

- Determination of the quadratic potential V (x);
- 2 Use of V (x) potential in the time-independent Schrödinger equation to obtain a dimensionless differential equation;
- Manipulation of the dimensionless differential equation to determine its solution's form at infinity;
- 4. Determination of the recursion relation that allows us to obtain complete solutions for the dimensionless differential equation;
- 5. Comparing successive power coefficients for the functions H (u) and e^U^2 quantization condition;
- 6. Writing the odd and even solutions using Hermite polynomials and writing the formulas for the energy eigenvalues;
- 7. Obtaining non-normalized Eigen functions;
- 8. Normalization of the Eigen functions and graph for levels 0, 1, 2, and 10.



Some trajectories of a harmonic oscillator according to Newton's laws of classical mechanics (A–B), and according to the Schrodinger equation of quantum mechanics (C–H). In A–B, the particle (represented as a ball attached to a spring) oscillates back and forth. In C–H, some solutions

to the Schrödinger Equation are shown, where the horizontal axis is position, and the vertical axis is the real part (blue) or imaginary part (red) of the wave function. C, D, E, F, but not G, H, are energy

Eigen states. H is a coherent state a quantum state that approximates the classical trajectory.

The Hamiltonian of the particle is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where m is the particle's mass, k is the

$$\omega = \sqrt{k/m}$$

is the angular frequency of the oscillator,

x^ is the position operator (given by x in the coordinate basis),

| $\hat{p}$ is the momentum operator

The first term in the Hamiltonian represents the kinetic energy of the particle, and the second term represents its potential energy, as in Hooke's law.

$$\langle\hat{x}^2\rangle = (2n+1)\frac{\hbar}{2m\omega} = \sigma_x^2$$

$$\langle\hat{p}^2\rangle = (2n+1)\frac{m\hbar\omega}{2} = \sigma_p^2$$

The functions H_n are Hermite polynomials

$$H_n(z) = (-1)^n\, e^{z^2}\frac{d^n}{dz^n}\left(e^{-z^2}\right)$$

The corresponding energy levels are

$$E_n = \hbar\omega\left(n+\tfrac{1}{2}\right) = (2n+1)\frac{\hbar}{2}\omega$$

The expectation values of position and momentum combined with variance of each variable can be derived from the wave function to understand the behavior of the energy Eigen Kets.

be $\langle \hat{x} \rangle = 0$ and $\langle \hat{p} \rangle = 0$ owing to the symmetry of the problem, whereas:

$$\langle \hat{x}^2 \rangle = (2n+1)\frac{\hbar}{2m\omega} = \sigma_x^2$$

$$\langle \hat{p}^2 \rangle = (2n+1)\frac{m\hbar\omega}{2} = \sigma_p^2$$

The variance in both position and momentum are observed to increase for higher energy levels. The lowest energy level has value of which minimum value due to uncertainty relation and also corresponds to a Gaussian wave function. The variance in both position and momentum are observed to increase for higher energy level

$$\sigma_x \sigma_p = \frac{\hbar}{2}$$

This energy spectrum is not worthy for three reasons. First , the energies are quantized, meaning that only discrete energy values (integer- plus- half multiples of ħω) are possible; this is a general feature of quant - mechanical systems when a particle is confined. Second, these discrete energy levels are equally spaced, unlike in the Bohr model of t atom, or the particle in a box. Third, the lowest achievable energy (the energy of the n = 0 state, called the ground state) is not equal to the minimum of the potential well, but ħω/2 above it ; this is called **zero- point energy**. Because of the zero- point energy, the position and moment um of the oscillator in the ground state are not fixed (as they would be in a classical oscillator), but have a small range of variance, in accordance with the Heisenberg uncertainty principle.

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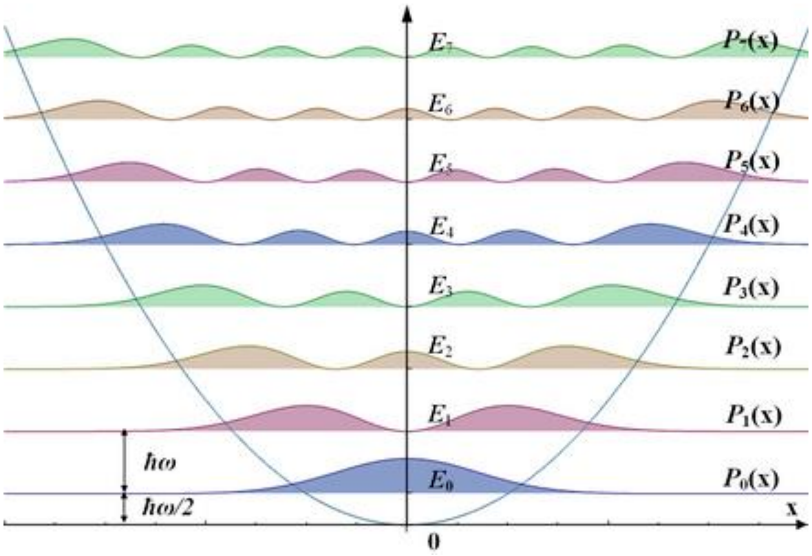


Some important axioms regarding quantum harmonic oscillator:

1. The ground state energy is always greater than zero which means a quantum oscillator is never at rest like its classical counterpart, even at the bottom of a potential well.

2. For high quantum numbers, the motion of a quantum oscillator becomes more similar to the motion of a classical oscillator as per the Bohr's correspondence principle.

3. The stationary states (states of definite energy) have nonzero values also in regions beyond classical turning points. When in the ground state, a quantum oscillator is most likely to be found around the position of the minimum of the potential well, which is the least-likely position for a classical oscillator.

4. The probability of finding a ground-state quantum particle in the classically forbidden region is about 16%.

5. Eigen-functions of quantum harmonic oscillator are nothing but the Hermite polynomials with degree depends on the oscillations.

6. Allowed energies are discrete and uniformly spaced for quantum harmonic oscillator and the spacing between the energies are as same as the Planck's energy quantum.

Solving the Schrodinger Equation

MATLAB m scripts can be used to find solutions of the Schrodinger Equation. The angular dependence is determined by evaluating the associated Legendre functions using the MATLAB function legendre(n, $\cos\theta$). The radial equation given by equation of allowed orbits be solved using the Matrix Method**.** The mscript qp_hydrogen.m can be used to solve the Schrodinger Equation for the hydrogen atom and hydrogen like ions. When the mscript qp_hydrogen. m is run, the following is shown requesting various inputs:

max radial distance (default 10e-10 m), **r_max** =
orbital quantum number (default 0), **L** = 0 magnetic quantum number (default 0),
m_L = 0 nuclear charge (default 1), **Z** = 1
Enter Principal Quantum Number for calculation of expectation values and graphical display
Enter Principal Quantum Number (n > L), **n** =

There are some problems with the accuracy of the Matrix Method due to the maximum range for the radial coordinate rmax. If rmax is too small than the energy eigenvalues near the top of the potential well will be inaccurate. However, if rmax is large, then the numerical procedure has difficulties in calculating the eigenvalues. The real potential diverges to infinity as r approaches zero. In our modeling, the potential energy function is truncated at some value of r. Table 1 gives the energy eigenvalues in eV for different values of rmax (x10-10 m). The theoretical values for E are calculated from quantized energy equation using the mscript qp_bohr.m .

Table 1

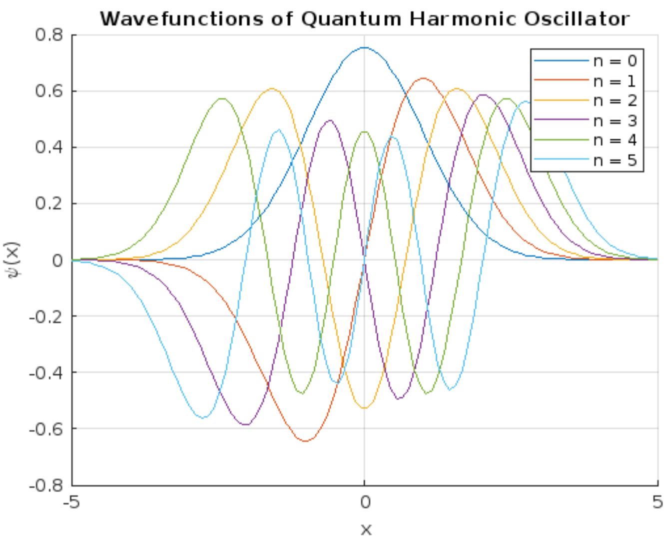
State ($n\ l\ m_l$)	E(theory)	r_{max} = 10	r_{max} = 20	r_{max} = 30	r_{max} = 50
(1 0 0)	-13.5828	-13.578	-13.586	-13.582	-13.5680
(2 0 0)	-3.3957	-3.3952	-3.3972	-3.3969	-3.3961
(3 0 0)	-1.5092	-1.2867	-1.5098	-1.5099	-1.5097
(4 0 0)	-0.8489	---	-0.81674	-0.84919	-0.84929
(5 0 0)	-0.5433	---	-0.21545	-0.52374	-0.54356
(6 0 0)	-0.377	---	---	-0.20996	-0.37605
(7 0 0)	-0.2772	---	---	---	-0.24948
(8 0 0)	-0.2122	---	---	---	-0.093909
(2 1 0)	-3.3957	-3.3980	-3.3984	-3.3985	-3.3986
(3 1 0)	-1.5092	-1.3537	-1.5103	-1.5104	-1.5104

MATLAB CODE (Quantum Harmonic Oscillator**)**


```
disp('Welcome to the Quantum Harmonic Oscillator Wavefunction Plotter!');
disp('--------------------------------------------------------------');
while true
    % Input quantum number for electron
    n_electron = input('Enter the quantum number for electron (non-negative integer), or type "exit" to quit: ', 's');
    if strcmpi(n_electron, 'exit')
        disp('Exiting program...');
        break;
    end
    n_electron = str2double(n_electron);
    % Check if the input is a non-negative integer
    if ~isnan(n_electron) && n_electron >= 0 && round(n_electron) == n_electron
        % Display the calculated formula
        disp(['Wavefunction formula for n_electron = ', num2str(n_electron), ':']);
        disp(['\psi_{electron}(x) = 1/sqrt(2^', num2str(n_electron), ' * ', 'factorial(', num2str(n_electron), ') * sqrt(pi)) * Hermite_', num

        % Define parameters
        x_max = 5; % Maximum x value for plotting
        num_points = 100; % Number of points for plotting
        % Define x values
        x = linspace(-x_max, x_max, num_points);
        % Calculate wavefunction for electron using Hermite polynomial
        Hn_electron = hermiteH(n_electron, x);
        psi_electron = 1/sqrt(2^n_electron * factorial(n_electron) * sqrt(pi)) .* Hn_electron .* exp(-x.^2/2);

        % Calculate probability density
        probability_density_electron = abs(psi_electron).^2;
        % Calculate wavefunction for proton using different quantum number
        n_proton = n_electron + 1; % Using a different quantum number for proton
        Hn_proton = hermiteH(n_proton, x);
        psi_proton = 1/sqrt(2^n_proton * factorial(n_proton) * sqrt(pi)) .* Hn_proton .* exp(-x.^2/2);

        % Calculate probability density
        probability_density_proton = abs(psi_proton).^2;
        % Create a figure and axis for animation
        figure;
        ax1 = subplot(2,1,1);
        line_handle_electron = plot(ax1, x, psi_electron, 'LineWidth', 1.5, 'Color', 'b');
        hold on;
        plot(ax1, x, probability_density_electron, '--', 'LineWidth', 1.5, 'Color', 'b'); % Plot probability density
        line_handle_proton = plot(ax1, x, psi_proton, 'LineWidth', 1.5, 'Color', 'r');
        plot(ax1, x, probability_density_proton, '--', 'LineWidth', 1.5, 'Color', 'r'); % Plot probability density
        xlabel(ax1, 'x');
        ylabel(ax1, '\psi(x), |\psi(x)|^2');
        title(ax1, ['Wavefunction and Probability Density of Quantum Harmonic Oscillator for n_electron = ', num2str(n_electron), ', n_proton =
        legend(ax1, {'\psi_{electron}(x)', '|\psi_{electron}(x)|^2', '\psi_{proton}(x)', '|\psi_{proton}(x)|^2'});
        grid(ax1, 'on');
       % Plot energy levels for electron
energies_electron = (0:n_electron) + 0.5;
ax2 = subplot(2,1,2);
for i = 1:length(energies_electron)
    if i == n_electron + 1 % Plot only the energy level corresponding to the entered quantum number
        line([x(1), x(end)], [energies_electron(i), energies_electron(i)], 'Color', 'b', 'LineStyle', '--', 'Parent', ax2);
        hold on;
    end
end
% Plot energy levels for proton
energies_proton = (0:n_proton) + 0.5;
for i = 1:length(energies_proton)
    if i == n_proton + 1 % Plot only the energy level corresponding to the quantum number of the proton
        line([x(1), x(end)], [energies_proton(i), energies_proton(i)], 'Color', 'r', 'LineStyle', '--', 'Parent', ax2);
        hold on;
    end
end
        % Animation loop
        t = 0;
        dt = 0.09; % Decreased pause duration for smoother animation
        while true
            % Update the wavefunctions with time-dependent phase factor
            psi_electron_t = psi_electron .* exp(-1i * t);
            psi_proton_t = psi_proton .* exp(-1i * t);
            set(line_handle_electron, 'YData', real(psi_electron_t));
            set(line_handle_proton, 'YData', real(psi_proton_t));
            drawnow;
            pause(dt); % Decreased pause duration

            % Increase time
            t = t + dt;

            % Check if user wants to stop animation
```

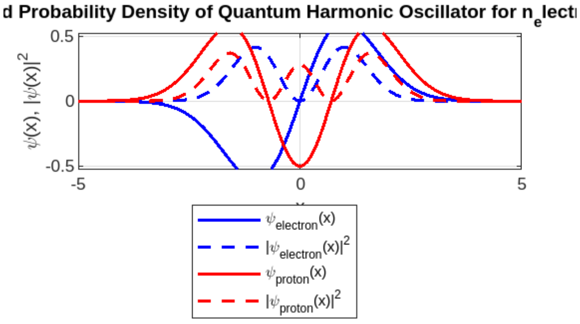
```
            if isempty(findobj('type','figure'))
                break;
            end
        end
        % Allow the user to adjust the range of the plot
        x_max = input('Enter the maximum x value for plotting (press Enter to continue with default): ');
        if ~isempty(x_max)
            x = linspace(-x_max, x_max, num_points);
            % Recalculate wavefunctions and probability densities
            Hn_electron = hermiteH(n_electron, x);
            psi_electron = 1/sqrt(2^n_electron * factorial(n_electron) * sqrt(pi)) .* Hn_electron .* exp(-x.^2/2);
            probability_density_electron = abs(psi_electron).^2;
            Hn_proton = hermiteH(n_proton, x);
            psi_proton = 1/sqrt(2^n_proton * factorial(n_proton) * sqrt(pi)) .* Hn_proton .* exp(-x.^2/2);
            probability_density_proton = abs(psi_proton).^2;
            % Update plots
            set(line_handle_electron, 'XData', x, 'YData', psi_electron);
            plot(ax1, x, probability_density_electron, '--', 'LineWidth', 1.5, 'Color', 'b');
            set(line_handle_proton, 'XData', x, 'YData', psi_proton);
            plot(ax1, x, probability_density_proton, '--', 'LineWidth', 1.5, 'Color', 'r');
            xlabel(ax1, 'x');
            ylabel(ax1, '\psi(x), |\psi(x)|^2');
            title(ax1, ['Wavefunction and Probability Density of Quantum Harmonic Oscillator for n_electron = ', num2str(n_electron), ', n_pro
            legend(ax1, {'\psi_{electron}(x)', '|\psi_{electron}(x)|^2', '\psi_{proton}(x)', '|\psi_{proton}(x)|^2'});
            grid(ax1, 'on');
        end
        input('Press Enter to continue...');
        close;
    else
        disp('Please enter a valid non-negative integer or type "exit" to quit.');
        continue; % Redirect the user to input the next quantum number
    end
end
```

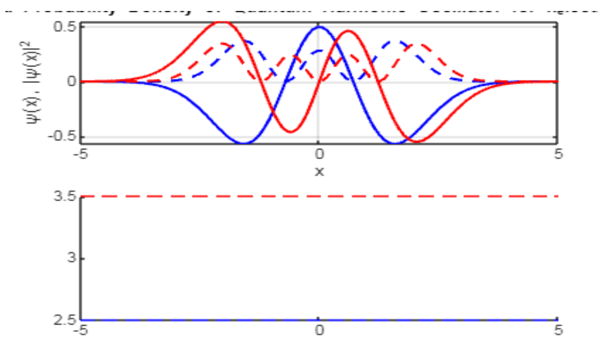
Generated Plots



The above plot shows the general nature of wave functions of qho for quantum numbers to 5 Similarly the wavefunctions for electron particle and proton particle can be seen in following figures generated using the MATLAB code I produced

For n=1 ***for n=2***

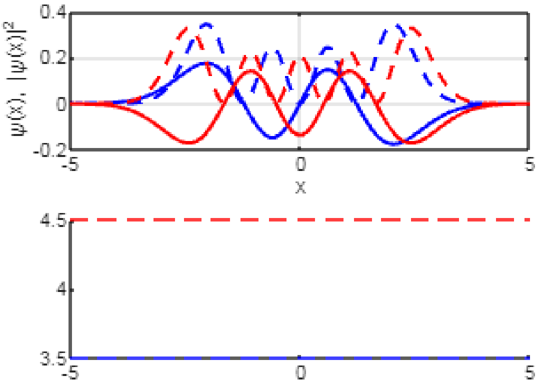




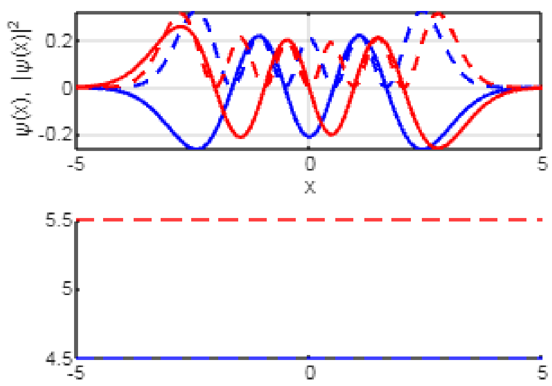
Wave function formula for n_electron = 1:
\psi_{electron}(x) = 1/sqrt(2^1 * factorial(1) * sqrt(pi)) * Hermite_1(x) * exp(-x^2/2)

Wave function formula for n_electron = 2:`\psi_{electron}(x) = 1/sqrt(2^2 * factorial(2) * sqrt(pi)) * Hermite_2(x) * exp(-x^2/2)`

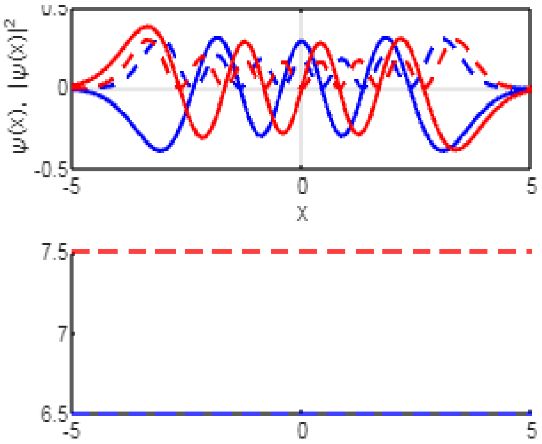
For n=3



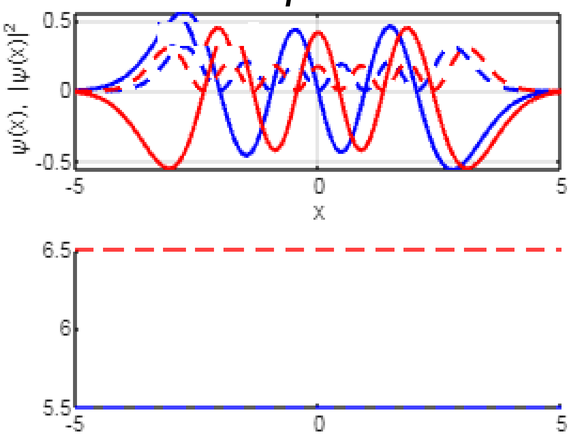
for n=4



For n=5



for n=6



Wave function formula for n_electron = 3:

\psi_{electron}(x) = 1/sqrt(2^3 * factorial(3) * sqrt(pi)) * Hermite_3(x) * exp(-x^2/2)

Wave function formula for n_electron = 4:

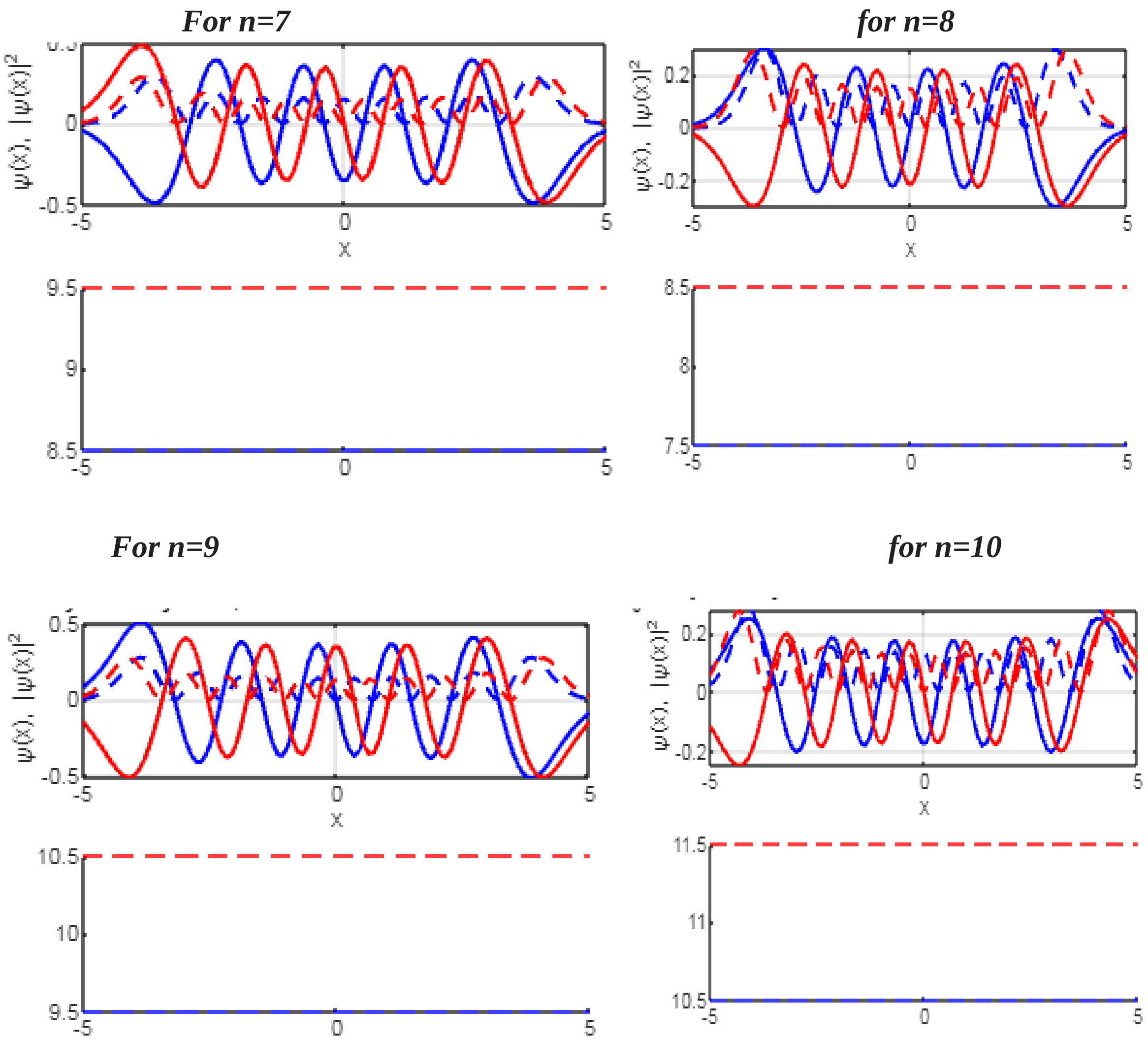
\psi_{electron}(x) = 1/sqrt(2^4 * factorial(4) * sqrt(pi)) * Hermite_4(x) * exp(-x^2/2

Wave function formula for n_electron = 5:

\psi_{electron}(x) = 1/sqrt(2^5 * factorial(5) * sqrt(pi)) * Hermite_5(x) * exp(-x^2/2)

Wave function formula for n_electron = 6:

\psi_{electron}(x) = 1/sqrt(2^6 * factorial(6) * sqrt(pi)) * Hermite_6(x) * exp(-x^2/2)



Wave function formula for n_electron = 7:

\psi_{electron}(x) = 1/sqrt(2^7 * factorial(7) * sqrt(pi)) * Hermite_7(x) * exp(-x^2/2)

Wave function formula for n_electron = 8:

\psi_{electron}(x) = 1/sqrt(2^8 * factorial(8) * sqrt(pi)) * Hermite_8(x) * exp(-x^2/2)

Wave function formula for n_electron = 9:

```
\psi_{electron}(x) = 1/sqrt(2^9 * factorial(9) * sqrt(pi)) * Hermite_9(x) * exp(-x^2/2)
```

Wave function formula for n_electron = 10:

```
\psi_{electron}(x) = 1/sqrt(2^10 * factorial(10) * sqrt(pi)) * Hermite_10(x) * exp(-x^2/2)
```

Plot Analysis QHO

Quantum Harmonic Oscillator

1. From above figures produced we can say that how is the wave functions graphs for n = 0, 1, 2, and up to 10.

2. The energy eigenvalues are expressed as a function of the force constant C and the particle mass m.

3. We have taken advantage of this to show the electron and proton wave functions simultaneously under the same potential V (x)

4. Meanwhile, Figures simultaneously presents the wave functions of levels n = 0, 1, 2, and 10 for electron (a) and proton (b) waves. In both, it is possible to identify the wave functions are evenly spaced.

5. Wave functions of levels n = 0, 1, 2, and 10 for (a) electron particle and (b) proton particle. In both cases, it is possible to identify that the wave functions of consecutive levels are evenly spaced.

Potential wells

Our goal is to find solutions of this form of the Schrodinger Equation for a potential energy function which traps the particle within a region. The negative slope of the potential energy function gives the force on the particle. For the particle to be bound the force acting on the particle is attractive. The solutions must also satisfy the boundary conditions for the wave function. The probability of finding the particle must be 1, therefore, the wave function must approach zero as the position from the trapped region increases. The imposition of the boundary conditions on the wave function results in a discrete set of values for the total energy *E* of the particle and a corresponding wave function for that energy, just like a vibrating guitar string which has a set of normal modes of vibration in which there is a harmonic sequence for the vibration frequencies.

The Schrodinger Equation can be solved analytically for only a few forms of the potential energy function. In this paper, we will consider a numerical approach to solving the Schrodinger Equation using a matrix method where the eigenvalues of a matrix gives the total energies of the particle and the Eigen functions the corresponding wave functions .Consider an electron in a potential well (see figure 1). For energy values below the top of the well, the physically acceptable solution of the time independent Schrodinger equation give a discrete set of energies which are the energy eigenvalues and corresponding to each eigenvalue there is the energy Eigen function. Quantization of the energy levels of bound particles arises naturally from the time independent Schrodinger equation and the boundary conditions imposed for physically acceptable solutions. The spectral lines observed in atomic systems are the result of transitions between such energy levels.

$$\frac{-\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+U(x)\psi(x)=E\psi(x)$$

$$\left[\frac{-\hbar^2}{2m}\frac{d^2}{dx^2}+U(x)\right]\psi(x)=E\psi(x)$$

$$H\psi(x)=E\psi(x)$$

where the H is the Hamiltonian operator, which is the operator that corresponds to the total energy of the system. This is an eigenvalue equation. The action of the operator H on the function returns the original function multiplied by a constant which could be complex. The eigenvalue equation is generally satisfied by a particular set of functions
and a corresponding set of constants $E1, E2$…… These are the
Eigen functions and the corresponding eigenvalues of the operator H.
The time independent Schrodinger Equation of a system is the energy eigenvalue equation of the system.

An Eigen function $\psi n[x]$ describes a state of define energy En. When the energy of the system in this state is measured, the result will always be En. For the Eigen function to represent physical sensible solutions, we require $\Psi n[x] \to 0$ as x-0 so that the wave function can be normalized.
For atomic systems it is more convenient to measure lengths in nm (nanometers) and energies in eV (electron volts).
We can use the scaling factors
length: Lse = 1×10-9 to convert m into nm
energy: Ese = 1.6×10-19 to convert J into eV.

$$\left[\left(\frac{-\hbar^2}{2m}\right)\left(\frac{1}{L_{se}^2 E_{se}}\right)\frac{d^2}{dx^2} + U(x)\right]\psi(x) = E\psi(x)$$

$$\left[C_{se}\frac{d^2}{dx^2} + U(x)\right]\psi(x) = E\psi(x) \quad \text{where} \quad C_{se} = \left(\frac{-\hbar^2}{2m}\right)\left(\frac{1}{L_{se}^2 E_{se}}\right)$$

$$H\psi(x) = E\psi(x)$$

Consider an electron in a potential well. For energy values below the top of the well, the physically acceptable solution of the time independent Schrodinger equation give a discrete set of energies which are the energy eigenvalues and corresponding to each eigenvalue there is the energy Eigen function. Quantization of the energy levels of bound particles arises naturally from the time independent Schrodinger equation and the boundary conditions imposed for physically acceptable solutions. The spectral lines observed in atomic systems are the result of transitions between such energy levels

Many problems in physics reduce to solving an eigenvalue equation, for example, the vibrations of a violin string. The eigenvalues and Eigen functions can be easily found using the MATLAB commands and scripts Eig thereby using the Matrix Method. To solve equation we first represent the continuous functions of x by sets of N discrete quantities expressed as vectors and matrices We can solve the Schrodinger Equation for a variety of different potential energy functions. The m-script Sewell_1s.m defines most of the constants and potential well parameters. Within the m-script you can change the code to modify the potential wells and add new potential wells. The first step in solving the Schrodinger Equation using the Matrix Method for a bound electron is to run the file Sewells_1.m and select the type of potential well.

The default potentials include:

1 Square well
2 Stepped well
3 Double well
4 Sloping well
5 Truncated Parabolic well
6 Morse Potential well
7 Parabolic fit to Morse Potential
8 Lattice

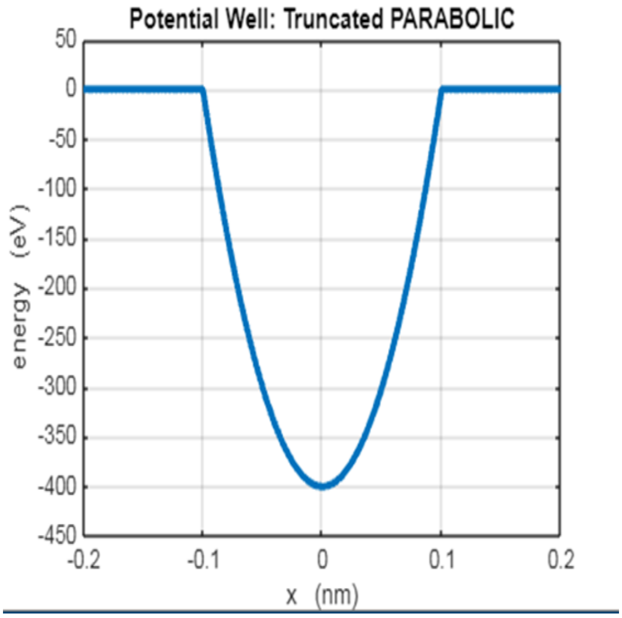
MATLAB Code For generating potential wells

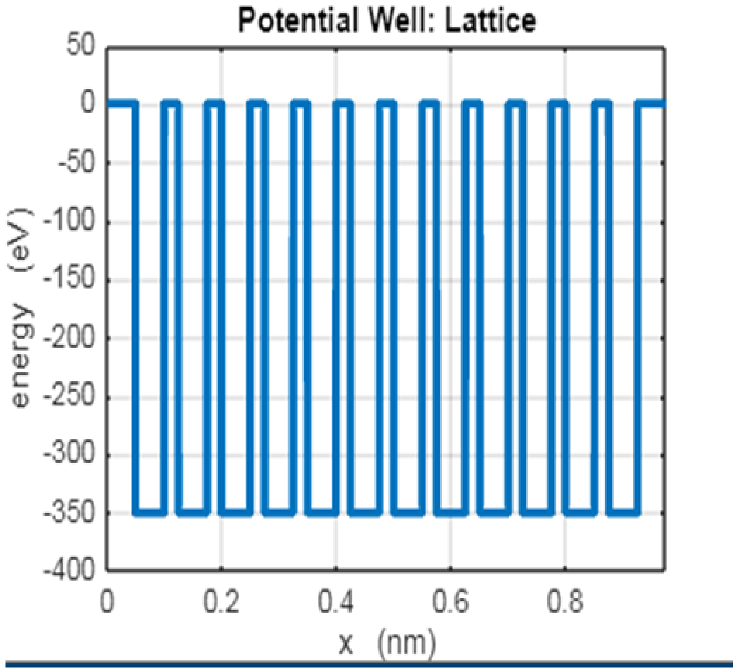
```
disp(' ');
inger Equation: Bound States');
set(gcf, 'NumberTitle', 'off');
set(gcf, 'PaperType', 'A4');
set(gcf, 'Units', 'normalized')
set(gcf, 'Position', [0.15 0.05 0.7 0.8]);
set(gcf, 'Color', [1 1 1]);
% Plotting wave fundisp(' ');
% Ensure the user inputs a valid quantum number
while true
   qn = input('Enter Quantum Number (1, 2, 3, ...), n: ');
   if qn >= 1 && qn <= length(E)
      break; % Valid input, exit loop
   else
      disp('Invalid input. Quantum number must be within the range.');
   end
end
% Compute Kinetic Energy (K) and Binding Energy (EB)
K = E(qn) - U;          % Kinetic energy (eV)
EB = -E(qn);            % Binding energy (eV)
% Plot the graphs
h_figure = figure(99);
clf
set(gcf, 'Name', 'Schrodinger equation and potential energy
axes('position', [0.1 0.6 0.35 0.32]);
[AX, H1, H2] = plotyyaxis(x, psi(:, qn), x, U);
title_m = [s, '   n = ', num2str(qn), '   E_n = ', num2str(E(qn))];
title(title_m, 'FontSize', 12);
xlabel('position x (nm)')
ylabel('wave function   psi');
set(get(AX(2), 'Ylabel'), 'String', 'U  (eV)')
set(AX(1), 'Xlim', [xMin - eps, xMax])
set(AX(1), 'YColor', [0 0 1])
set(AX(2), 'Xlim', [xMin - eps, xMax])
set(AX(2), 'Ylim', [min(U) - 50, max(K) + 50])
set(AX(2), 'YColor', [1 0 0])
set(H1, 'color', [0 0 1], 'LineWidth', 3);
set(H2, 'color', [1 0 0]);
line([xMin, xMax], [-EB, -EB], 'LineWidth', 2, 'Color', [0 0 0], 'Parent', AX(2))
line([xMin, xMax], [0, 0], 'Color', [0 0 1], 'Parent', AX(1))
% Plotting probability density
axes('position', [0.1, 0.2, 0.8, 0.32]);
[AX, H1, H2] = plotyyaxis(x, prob(:, qn), x, U);
xlabel('position x (nm)')
ylabel('prob density   psi?');
set(get(AX(2), 'Ylabel'), 'String', 'U  (eV)')
set(AX(1), 'Xlim', [xMin - eps, xMax])
set(AX(1), 'YColor', [0 0 1])
```

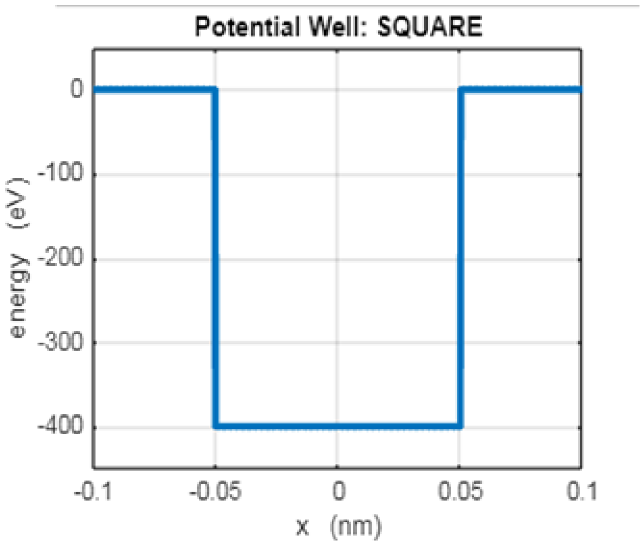
```
set(AX(2), 'Xlim', [xMin - eps, xMax])


set(AX(2), 'Ylim', [min(U) - 50, max(U) + 50])
set(AX(2), 'YColor', [1 0 0])
set(H1, 'color', [0 0 1], 'LineWidth', 3);
set(H2, 'color', [1 0 0]);
line([xMin, xMax], [-EB, -EB], 'LineWidth', 2, 'Color', [0 0 0], 'Parent', AX(2))
% Plotting potential energy, kinetic energy, and total energy
axes('position', [0.57, 0.6, 0.33, 0.32]);
plot(x, U, 'r', 'LineWidth', 2);
hold on
plot(x(1:75:end), K(1:75:end), 'g+');
plot(x, K, 'g');
title('E_n(black)   U(red)    K(green)');
xlabel('position x (nm)');
ylabel('energies  (eV)');
line([xMin, xMax], [-EB, -EB], 'LineWidth', 2, 'Color', [0 0 0]);
set(gca, 'Xlim', [xMin, xMax]);
set(gca, 'Ylim', [min(U) - 50, max(K) + 50]);
% Plotting particle location after measurement
axes('position', [0.1, 0.05, 0.8, 0.08]);
hold on
num1 = 10000;
axis off
for c = 1:num1
   xIndex = ceil(1 + (num - 2) * rand(1, 1));
   yIndex = rand(1, 1);
   pIndex = max(prob(:, qn)) * rand(1, 1);
   if pIndex < prob(xIndex, qn)
      plot(x(xIndex),  yIndex,  's',  'MarkerSize',  2,  'MarkerFaceColor',  'b',  'MarkerEdgeColor',
'none');
   end
end
set(gca, 'Xlim', [xMin, xMax]);
set(gca, 'Ylim', [0, 1]);

hold off
```

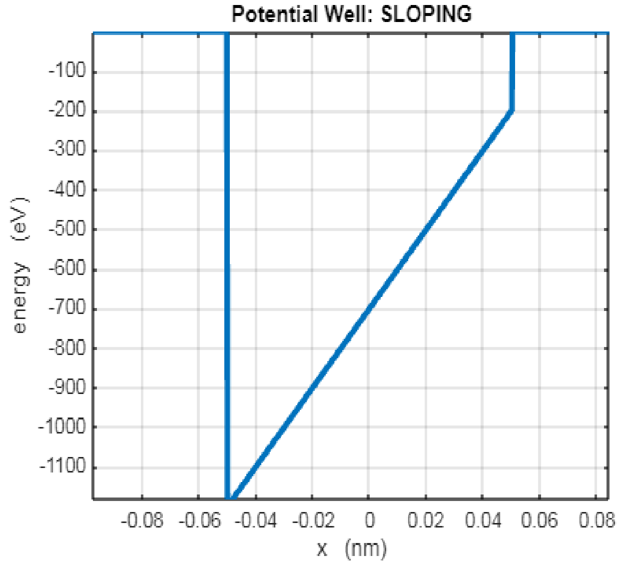
Results

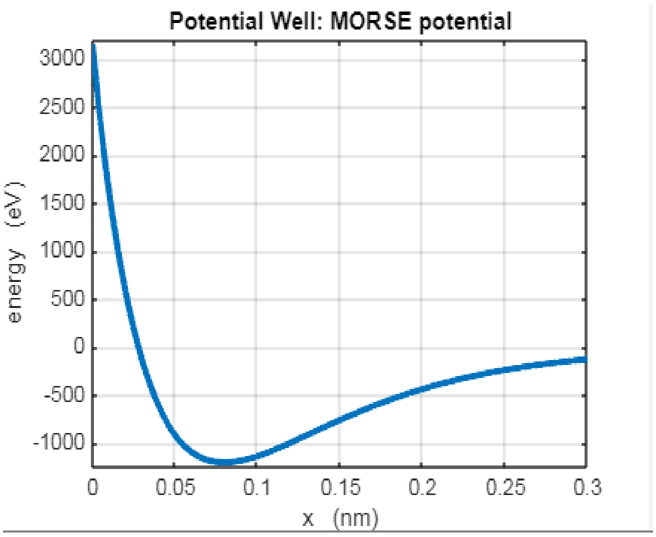
Some of the default potential wells that can be generated using the m-script wells3.m User can change any of the parameters in the m-script se_wells3.m and run it again. Then to solve the Schrodinger Equation run the m-script se_solve.m.

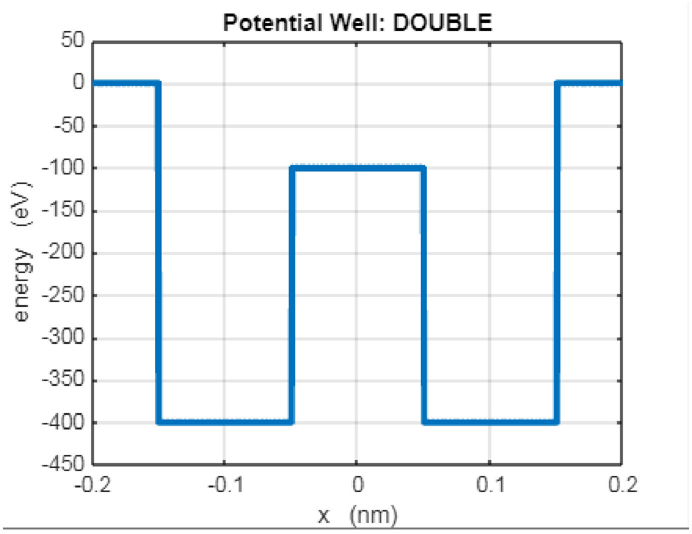


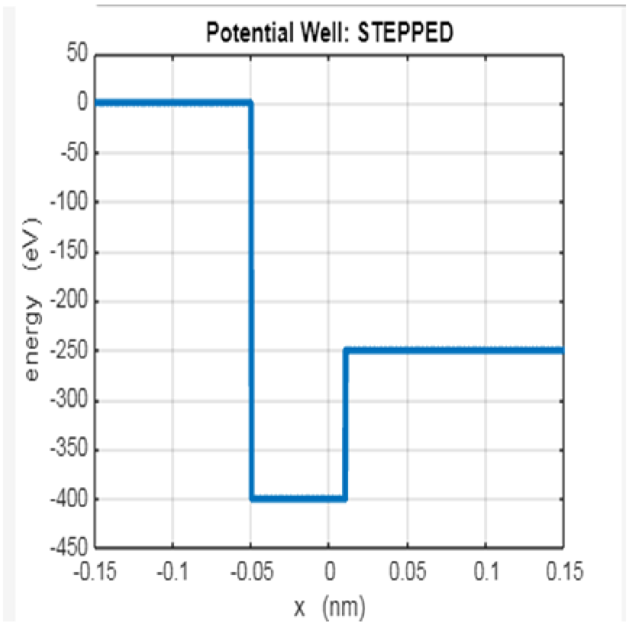



Potential Well: SQUARE
energy (eV)
x (nm)

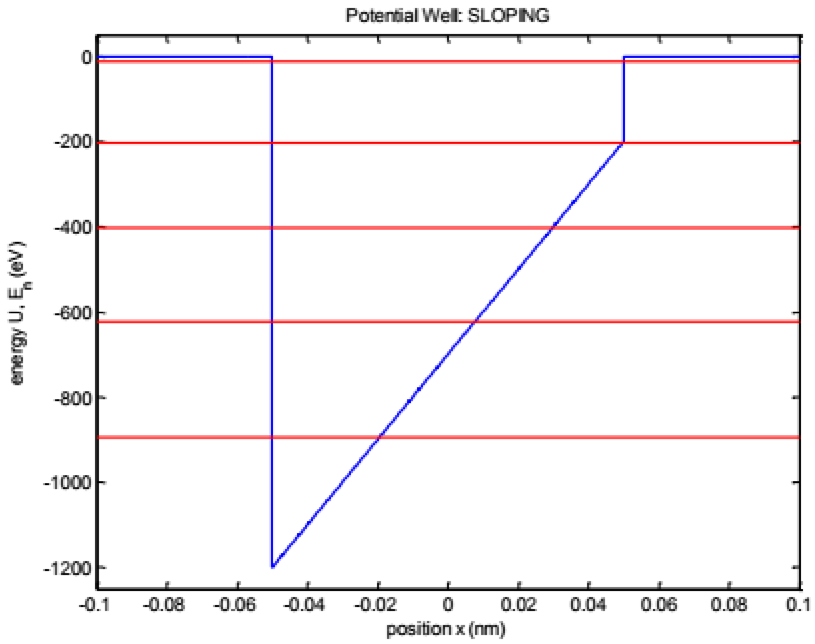

Potential Well: SLOPING
energy (eV)
x (nm)


Potential Well: MORSE potential
energy (eV)
x (nm)

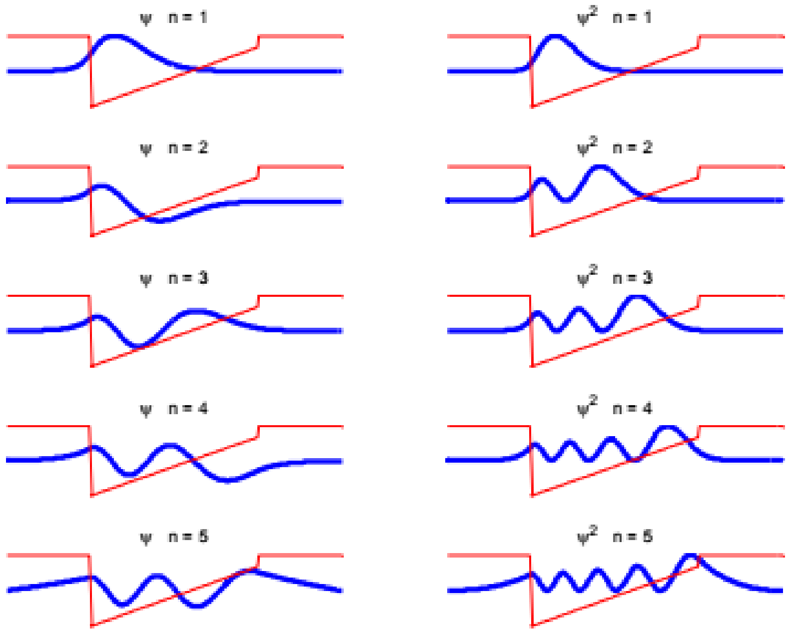

Potential Well: DOUBLE
energy (eV)
x (nm)


Potential Well: STEPPED
energy (eV)
x (nm)

A graph of the energy spectrum for a sloping potential well



The energy eigenvectors and probability distributions for sloping potential well



Conclusions

1. It is possible to simulate and plot the quantum harmonic oscillator and its other applications.

2. The potential wells here can be generated by user by entering the appropriate parameters.

3. The Eigen vectors and Eigen values and the energy spectrum for each potential well can be plotted for a better clarity.

4. Just like Hydrogen atom the Schrodinger equation for more than one electron system can be solved and the probability densities can be plotted or simulated

.

Future Scope

1. The similar type of modeling can be done for more than one electron system such as lithium and Helium and their ions that can be done by using other languages such as Java-script and python.

2. Similarly it the above project can be further extended by taking into account the different types of potential wells and investigating their solutions per well.

3. The open source quantum mechanical platforms such as IBM Quiskit can be used to solve modern logic gates problems as well

References

1. **http s://link.springer. com /article/1 0 .1 0 0 7 /s40 5 09 - 0 21 -00 2 50- 0**

2. **http s://en.wikipedia.org /wiki/ Quantum harmonic oscillator**

3. **http s://in. mathworks.com /matlab central /file exchange/5 1 9 9 0 - excited - states-of -quantum –harmonic oscillator-using -raising - operator**

4. MATLAB simulations by Dan green.

5.www.Mathworks.com.

6. www.source.github.com .

7.Quantum mechanics By Dj Griffiths

Comparison of results with calculations

	HBohr		H simulation	
n	$\boldsymbol{E_B}$**(eV)**	$\boldsymbol{r_{n(*10^{-10})m}}$	$\boldsymbol{E_B}$**(eV)**	$\boldsymbol{r_{n(*10^{-10})m}}$
1	13.605	0.5292	13.559	0.550
2	3.401	2.112	3.400	2.110
3	1.512	4.763	1.510	4.750
4	0.850	8.467	0.849	8.500
5	0.5442	13.2295	0.544	13.25
6	0.3779	19.0505	0.377	19.05
7	0.2776	25.9299	0.273	25.95
8	0.2126	33.8676	0.175	33.45
9	0.1680	42.8637	0.049	39.65
10	0.1360	52.9182	---	43.45

There is excellent agreement between the Bohr Theory predictions and those of the simulation up to about n = 7. For higher n values, inaccuracies occur because the forcing of the wave function to go to zero at a radial distance of rmax = 60x10-10 m, the maximum well was set at -1000 eV .